Mathematical Methods — Units 3 & 4

Differentiation · Practice Examination 1 (Technology-free)

Instructions: Answer all questions in the spaces provided. Show all working. Total: 20 marks · Writing time: 30 minutes.

1. Differentiate $y = x^3 - 4x^2 + 7x - 2$. (3 marks)

2. Find dy/dx for $y = (2x + 1)^5$. (3 marks)

3. Find the equation of the tangent to $y = x^2 - 3x$ at the point where $x = 2$. (4 marks)

4. Differentiate $y = x\,e^x$ and find the x-coordinate of the stationary point. (3 marks)

5. Differentiate $y = \ln(3x^2 + 1)$. (3 marks)

6. For the curve $y = x^3 - 6x^2 + 9x$, find the coordinates and nature of the stationary points. (4 marks)